

# Polynomial Regression Assignment - IMT2024031

## Final report

This report describes the polynomial-regression models used for the two assigned datasets. The model-selection criterion is 5-fold cross-validation mean squared error (MSE), with R-squared reported as a secondary metric.

| Problem | Inputs | Degree search | Selected degree | Terms | CV MSE   | CV R <sup>2</sup> |
|---------|--------|---------------|-----------------|-------|----------|-------------------|
| var1    | 6      | 1-10          | 4               | 210   | 0.873890 | 0.907736          |
| var2    | 3      | 1-20          | 8               | 165   | 0.251895 | 0.994196          |

## 1. Methodology

For each problem, a multivariate polynomial basis was generated with PolynomialFeatures. The degree was searched over the full range allowed by the assignment: 1 through 10 for var1 and 1 through 20 for var2. For every degree, a 5-fold shuffled cross-validation split (random state 42) was used. Each fold fit ordinary least-squares linear regression on the polynomial features. The degree with the lowest mean validation MSE was selected and then the corresponding polynomial model was retrained on all available training samples.

The final models therefore use polynomial regression only. No Ridge/Lasso regularization, tree model, neural network, or other nonlinear estimator is used. The supplied features are already scaled approximately to  $[-1, 1]$ , which also keeps the lower-degree polynomial basis numerically well behaved.

## 2. Model selection rationale

For var1, validation error decreases through degree 4 and then rises substantially at degree 5 and above under ordinary least squares. Degree 4 gives the minimum 5-fold CV MSE of 0.873890 and a mean CV R<sup>2</sup> of 0.907736. The selected degree 4 basis contains 210 terms including the intercept.

For var2, validation error falls rapidly up to degree 8, reaches a minimum there, and then begins to increase. Higher degrees become increasingly unstable, with very large cross-validation errors from degree 14 onward. Degree 8 gives the minimum 5-fold CV MSE of 0.251895 and mean CV R<sup>2</sup> of 0.994196. The selected degree 8 basis contains 165 terms including the intercept.

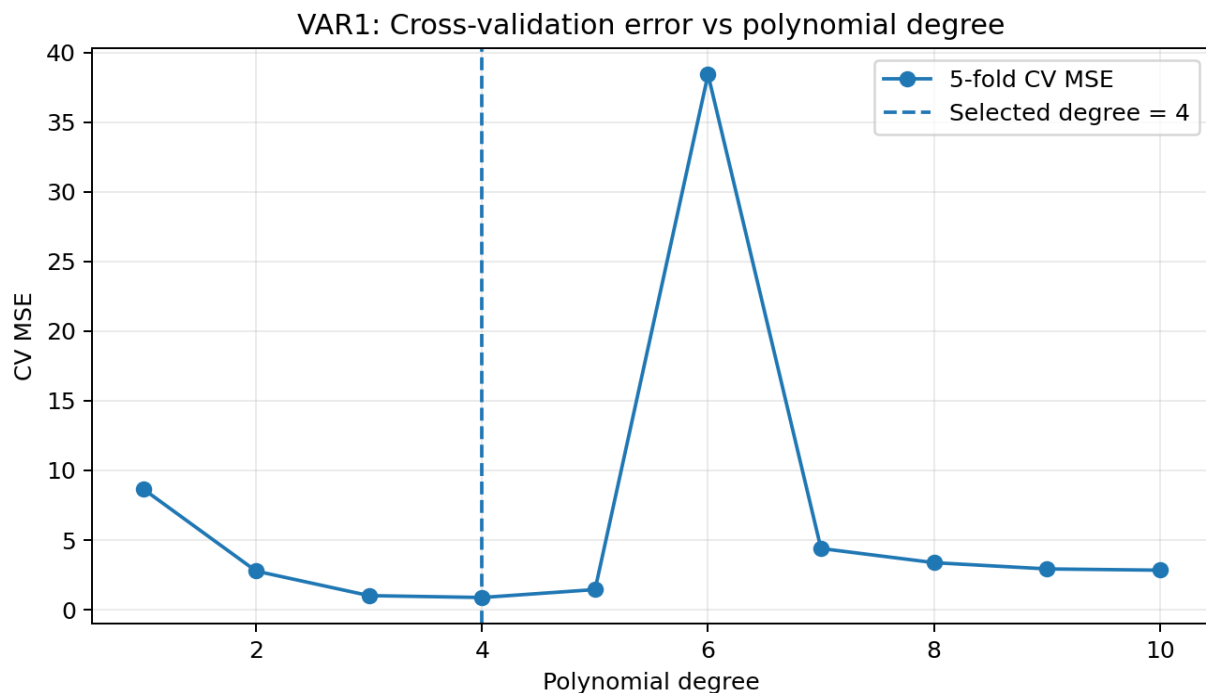


Figure 1. var1 degree selection using exhaustive 5-fold cross-validation.

### 3. var1: Power Plant Steam Turbine Optimization

The training set contains 1000 observations with six input variables (x1-x6) and target y. The candidate polynomial degree was evaluated from 1 through 10. The selected degree is 4. This corresponds to 210 multivariate monomials including the constant term.

| Degree | Terms | CV MSE    | CV R <sup>2</sup> |
|--------|-------|-----------|-------------------|
| 1      | 7     | 8.665146  | 0.092537          |
| 2      | 28    | 2.768877  | 0.708216          |
| 3      | 84    | 1.008131  | 0.893781          |
| 4      | 210   | 0.873890  | 0.907736          |
| 5      | 462   | 1.443936  | 0.846210          |
| 6      | 924   | 38.426435 | -3.073894         |
| 7      | 1716  | 4.392951  | 0.538595          |
| 8      | 3003  | 3.376354  | 0.646055          |
| 9      | 5005  | 2.926298  | 0.692701          |
| 10     | 8008  | 2.834769  | 0.702120          |

The degree-4 model is used to generate the 1000 predictions in IMT2024031\_pred\_var1.csv. Because test-set ground truth is hidden, the report does not claim a test MSE or test R<sup>2</sup> for the submitted predictions.

## 4. var2: Subterranean Thermal Reservoir Mapping

The training set contains 1000 observations with three spatial coordinates (x1-x3) and target y. The candidate polynomial degree was evaluated from 1 through 20. The selected degree is 8, with 165 polynomial terms including the constant term.

| Degree | Terms | CV MSE      | CV R <sup>2</sup>     |
|--------|-------|-------------|-----------------------|
| 1      | 4     | 32.7296     | 0.253667              |
| 2      | 10    | 21.1709     | 0.515386              |
| 3      | 20    | 11.2736     | 0.739961              |
| 4      | 35    | 3.53035     | 0.918520              |
| 5      | 56    | 1.46047     | 0.966362              |
| 6      | 84    | 0.584596    | 0.986529              |
| 7      | 120   | 0.312781    | 0.992757              |
| 8      | 165   | 0.251895    | 0.994196              |
| 9      | 220   | 0.276391    | 0.993686              |
| 10     | 286   | 0.567752    | 0.987805              |
| 11     | 364   | 0.838317    | 0.981036              |
| 12     | 455   | 1.63891     | 0.960434              |
| 13     | 560   | 13.463      | 0.682398              |
| 14     | 680   | 475.857     | -10.311305            |
| 15     | 816   | 3.05081e+12 | -67389366767.338516   |
| 16     | 969   | 8.98334e+12 | -170823025568.581635  |
| 17     | 1140  | 3.08428e+14 | -5879150284734.886719 |
| 18     | 1330  | 1.50843e+14 | -3128991421484.368164 |
| 19     | 1540  | 7.80959e+13 | -1693874965702.667480 |
| 20     | 1771  | 6.00631e+13 | -1325010717282.423096 |

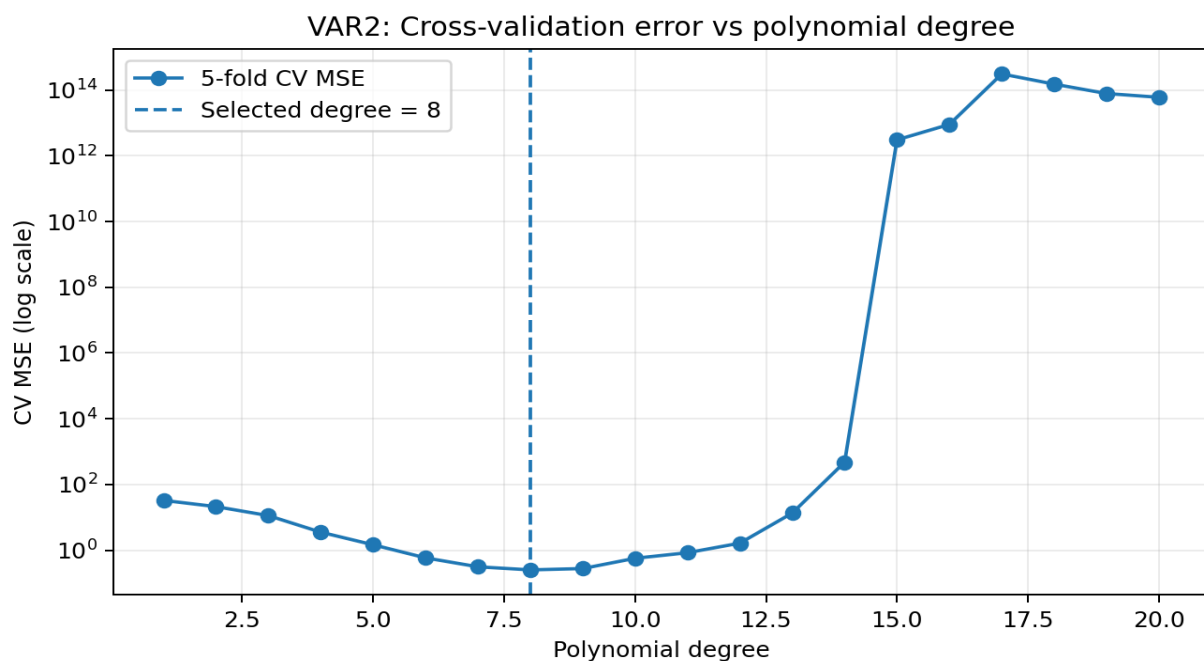


Figure 2. var2 degree selection. The logarithmic vertical axis makes the severe high-degree instability visible.

## 5. Final outputs and reproducibility

The final prediction files each contain exactly one column named `y` and 1000 rows, matching the supplied test-set size. The Python source performs the full degree search, saves the complete CV tables, retrains the selected ordinary polynomial regression on all training data, and writes the two prediction files. The ZIP package is organized so that the code and report can be uploaded to a GitHub repository as required by the assignment.